Liebherr Group – Final speaker notes

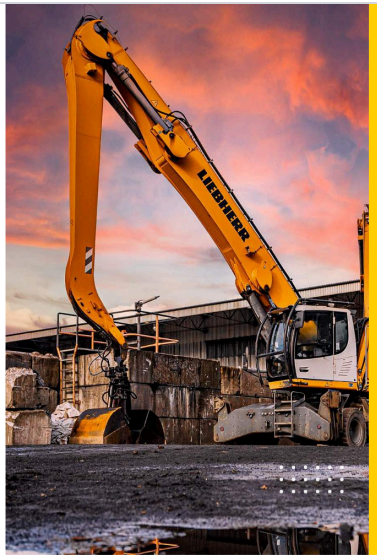
HAT1 Financial Reporting · 12 talk slides · planned speaking time ≈ 9:50 min (1251 words at ≈ 125 words per minute) · backup slides 13–44 for the Q&A

Central question: **How financially strong is Liebherr, and which factors will drive its future performance? For a long-term investor: take a minority stake now – or not yet?**

SPEAKER	SLIDES	TOPICS	TIME	HANDOVER LINE
Adam	1–3	Title; Executive summary; Company and environment	**2:15**	So how does Liebherr compete in this market? Zarah will now show you its competitive advantage, and what it means for the numbers.
Zarah	4–5	Competitive advantage; Financial performance	**2:15**	But can we trust the reported margin? Philipp will now look behind the figures.
Philipp	6–8	Accounting analysis – judgements; Accounting adjustments vs. sensitivities; Financial position and cash flow	**2:50**	Simon will now show you where the cash is tied up, and what all of this means for the investor.
Simon	9–12	Working capital; Investment assessment; Conclusion; Thank you	**2:30**	Closes the talk and opens the Q&A

→ HANDOVER last sentence before the next speaker · **● CLICK** slide 10: one click per condition (3 clicks), then one click to slide 11 · In the HTML version press **P** for the presenter view and **I** for the backup index.

01 **Adam** **Title** 0:35



HAT1 FINANCIAL REPORTING · 2026

Liebherr Group

Financial Reporting and Investment Analysis 2022–2025

CENTRAL QUESTION

How financially strong is Liebherr, and which factors will drive its future performance?

For a long-term investor: take a minority stake now – or not yet?

Benchmark: Caterpillar Inc. and the average of 105 listed US machinery companies

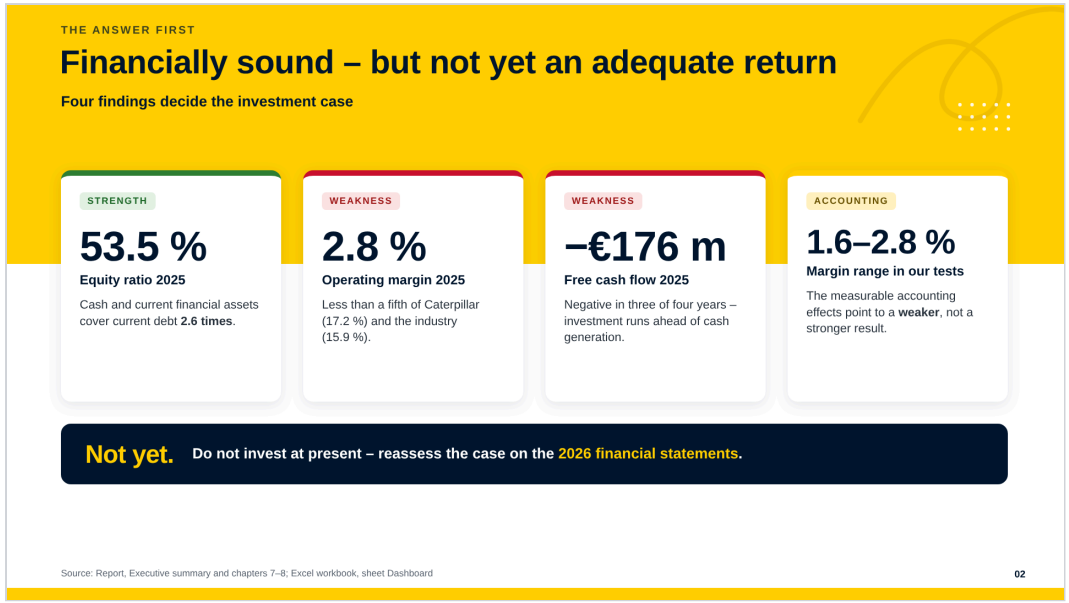
PRESENTED BY

Simon Beermann · Adam Fliegel · Zofia Narkiewicz · Philipp Hoffmann · Zarah Gilmore Islam

Fontys University of Applied Sciences, Venlo · International Finance and Control · Lecturer: Kim Verspeek

Good morning everyone. We are Simon, Adam, Zofia, Philipp and Zarah, and today we present our analysis of the Liebherr Group from 2022 to 2025. My name is Adam, and I will start.

Our report is written for a long-term investor who is considering a minority stake in Liebherr. Our central question is how financially strong Liebherr is, and which factors will drive its future performance. For the investor, the decision is simple. Invest now, or not yet?

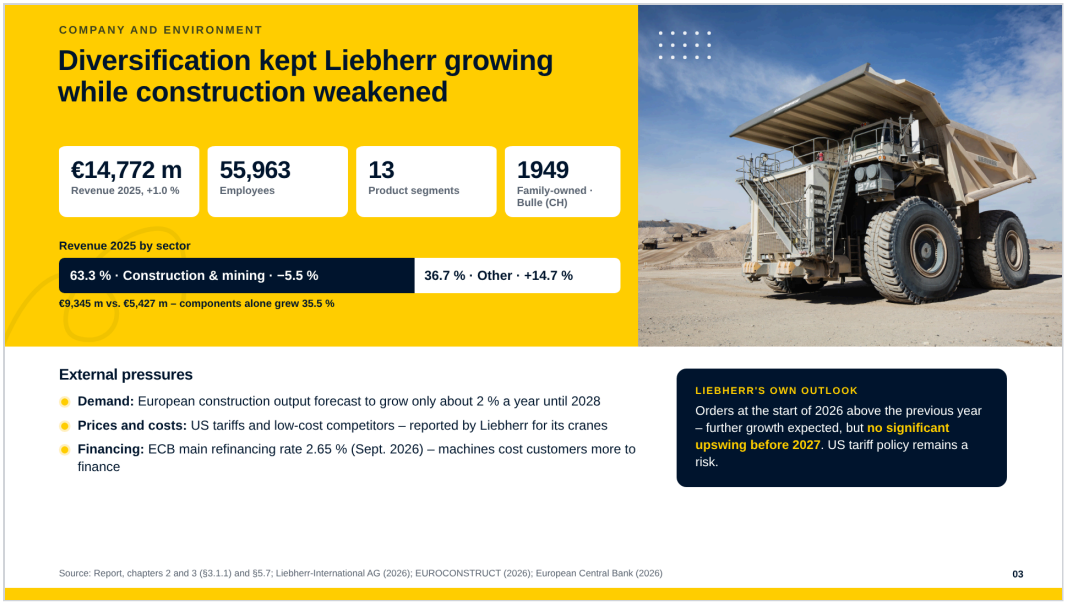


We want to give you the answer first, and our answer is not yet.

Liebherr is financially sound. Equity finances 53.5 per cent of total assets, and cash covers current debt 2.6 times. So the risk of financial distress is low.

But a strong balance sheet is not a good investment by itself. The operating margin was only 2.8 per cent. That is less than a fifth of Caterpillar and of the industry. And free cash flow was negative in three of the four years.

So our advice is to wait, and to reassess the case on the 2026 statements. Let us show you why.



Liebherr is a family-owned, unlisted group with almost 56,000 employees in 13 product segments.

In 2025 construction and mining, which make up almost two thirds of revenue, fell by 5.5 per cent. The other segments grew by 14.7 per cent, and that is why the group still grew a little.

The environment is difficult. Construction in Europe grows only slowly, US tariffs and cheap competitors press on prices, and higher interest rates make machines more expensive. Liebherr itself does not expect a real upswing before 2027.

→ HANDOVER So how does Liebherr compete in this market? Zarah will now show you its competitive advantage, and what it means for the numbers.



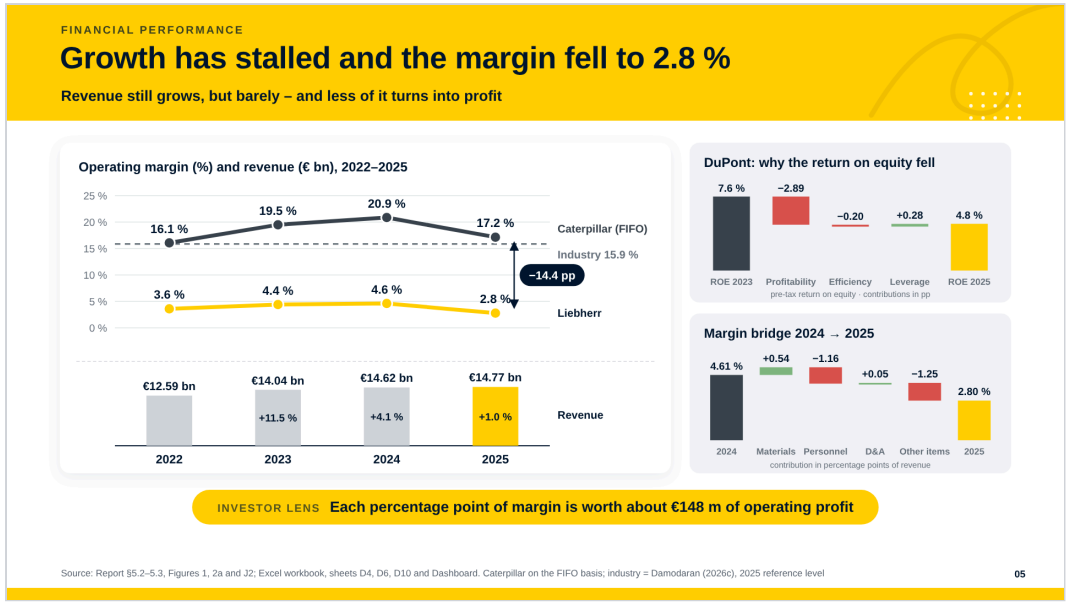
Thank you, Adam.

Three advantages support Liebherr's strategy.

The first one is diversification. Thirteen segments spread the risk, and in 2025 the other segments grew while construction fell. The second one is vertical integration. Liebherr builds key technologies itself and sells through its own companies instead of dealers. The third one is innovation. Research and development rose every year to 708 million euro, a higher share of revenue than at Caterpillar.

But each advantage has a price. Diversification does not raise the margin, vertical integration ties up capital, and research spending kept rising while profit fell.

So the advantage is real, but it sits in the asset base and not yet in the margin. Does it turn into returns? Let's look at the numbers.



The bars at the bottom show revenue. It grew by 11.5 per cent in 2023, but only by 1 per cent in 2025.

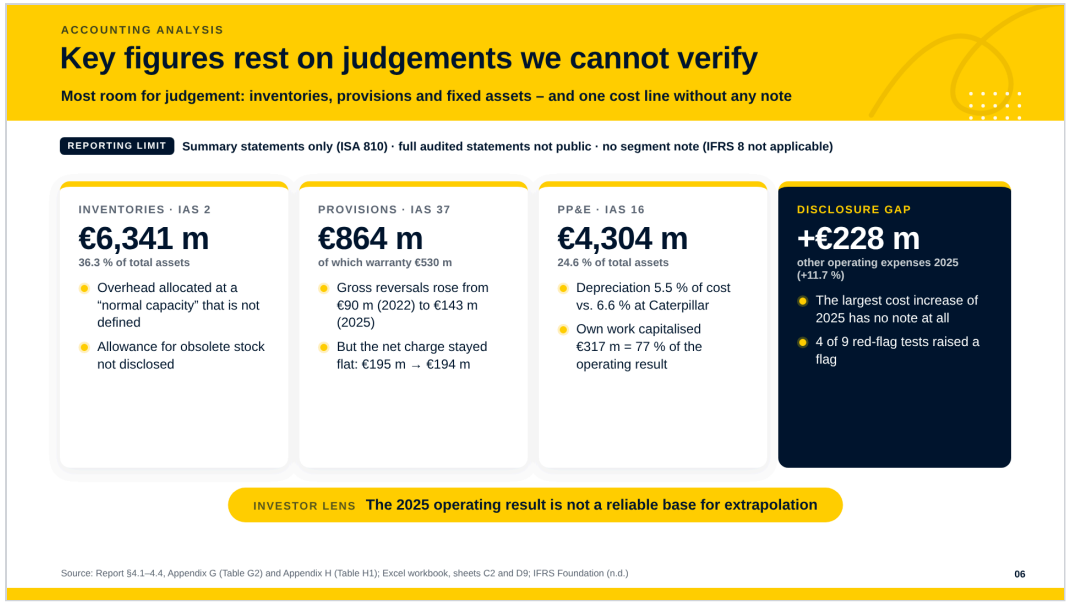
The lines above show the bigger problem. Liebherr's margin, in yellow, was around 4 per cent and fell to 2.8 per cent in 2025. Caterpillar earned 16 to 21 per cent. The gap was there in every single year, so this is not just one bad year.

The bridge on the right explains 2025. Personnel costs grew five times faster than revenue, and other operating items cost another 1.25 points, without any explanation.

The DuPont analysis shows what this means for the owners. The pre-tax return on equity fell from 7.6 to 4.8 per cent, and almost all of it comes from lower profitability, not from debt.

Each point of margin is worth about 148 million euro.

→ HANDOVER But can we trust the reported margin? Philipp will now look behind the figures.

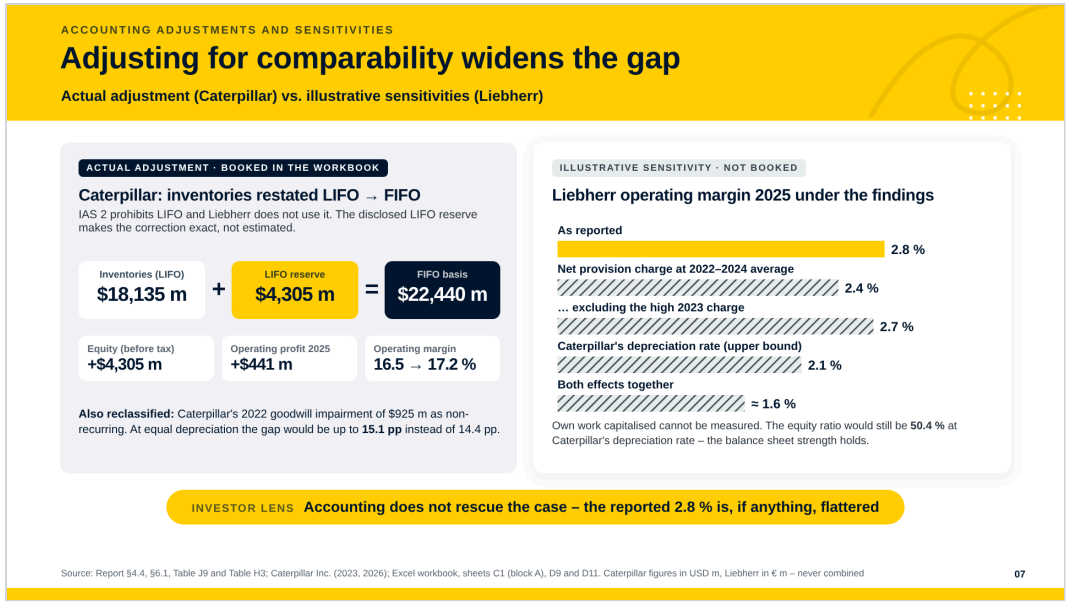


Thank you, Zarah.

Liebherr publishes only summary statements. The full audited statements are not public, and there is no segment note.

Three accounts carry most of the judgement. For inventories, the allowance for old stock is not disclosed. For provisions, the reversals rose to 143 million euro, but the net charge stayed flat. And for fixed assets, Liebherr depreciates more slowly than Caterpillar.

The fourth box matters most. Other operating expenses rose by 228 million euro, and there is no note at all. So the biggest cost increase of 2025 cannot be explained. Four of our nine red-flag tests raised a flag.

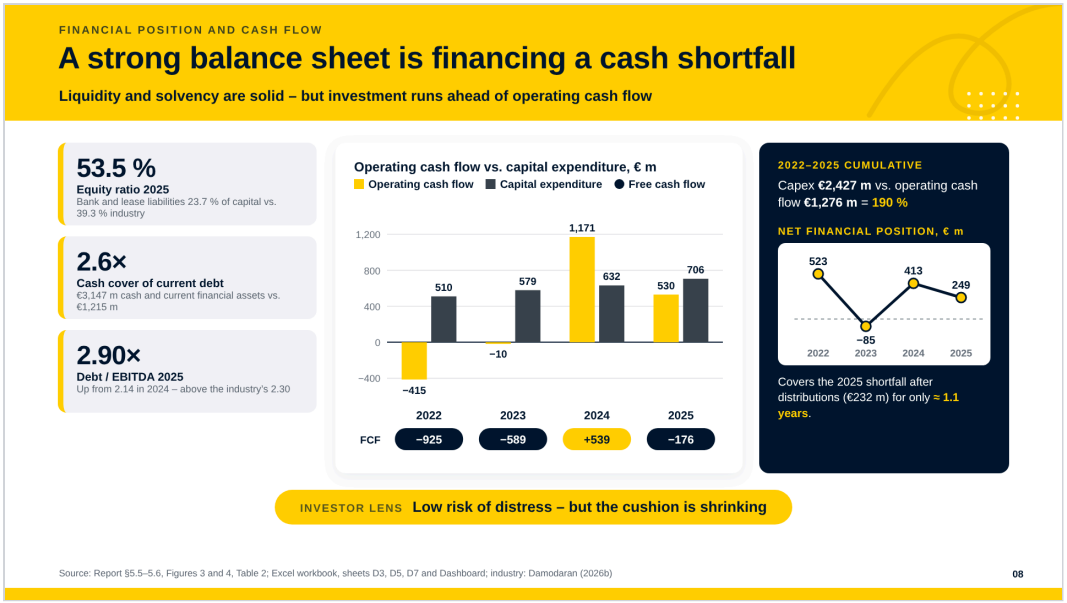


On the left you see an actual adjustment that we made. On the right you see sensitivities that we only measured.

Caterpillar values most of its inventory with LIFO, which IAS 2 does not allow. Because Caterpillar discloses its LIFO reserve, the correction is exact, and it adds 441 million dollars of operating profit. Caterpillar's margin rises from 16.5 to 17.2 per cent, so the gap gets bigger, not smaller.

For Liebherr we only illustrate. With a normal provision charge the margin would be 2.4 per cent, and if we also use Caterpillar's depreciation rate, about 1.6 per cent.

So the accounting does not hide a better company. But the balance sheet strength survives, because the equity ratio stays at around 50 per cent.

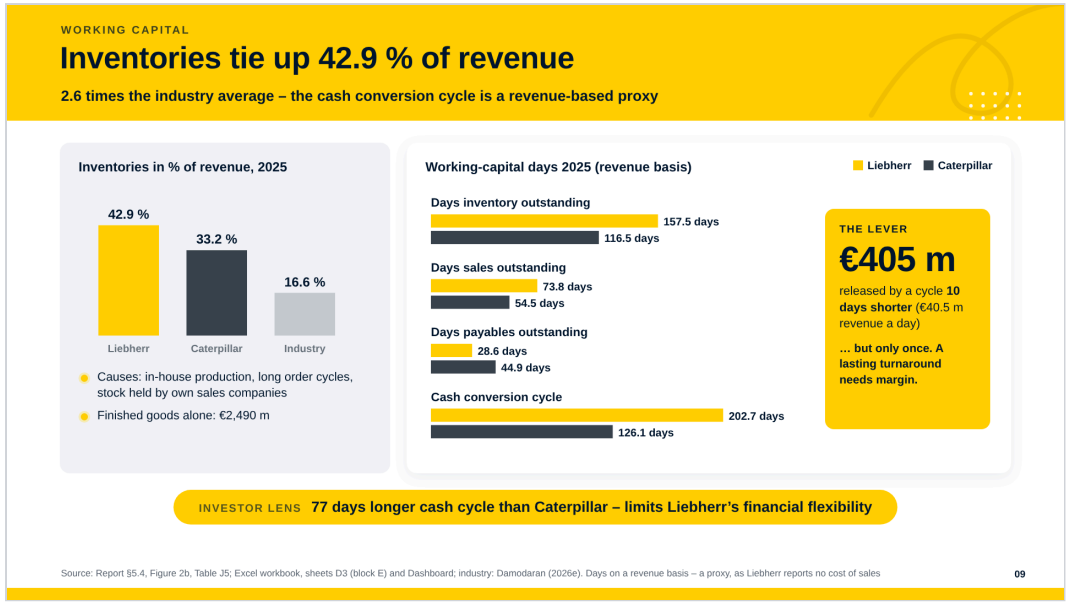


The balance sheet is Liebherr's clearest strength. But now let's look at the cash.

The yellow bars are operating cash flow, and the grey bars are capital expenditure. Over four years Liebherr invested 2.4 billion euro, but generated only 1.3 billion from operations. That is 190 per cent. Free cash flow was negative in three of the four years.

As a result, the cushion is shrinking. The net financial position fell from 523 to 249 million euro. If 2025 happened again, the cushion would last only about one more year. The balance sheet buys time, but it does not replace returns.

→ HANDOVER Simon will now show you where the cash is tied up, and what all of this means for the investor.

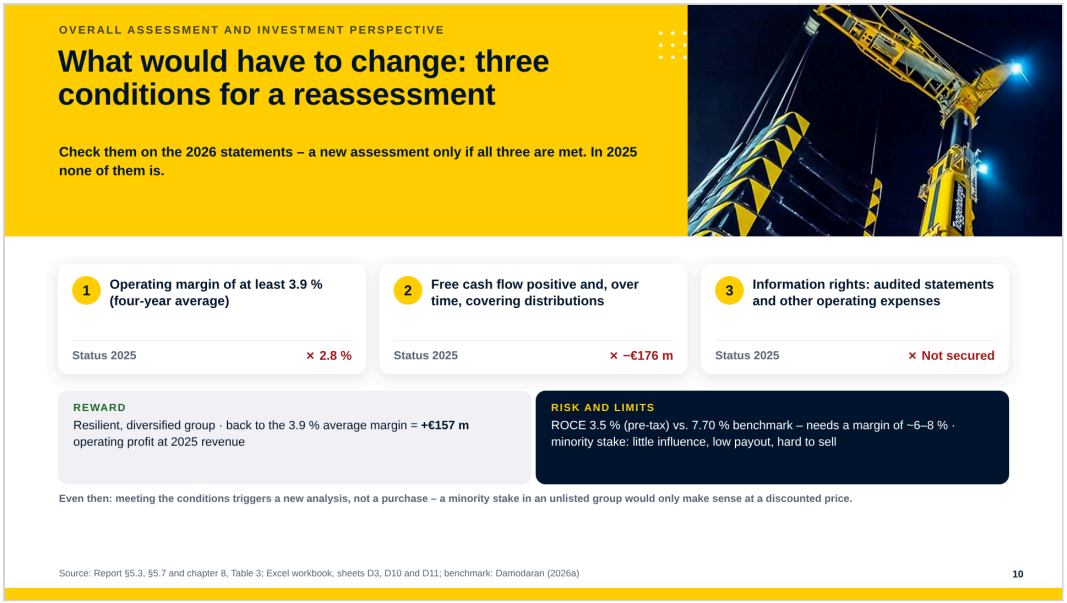


Thank you, Philipp.

So where is the cash tied up? Mostly in inventories. They equal 42.9 per cent of revenue, which is 2.6 times the industry.

Measured in days of revenue, the cash conversion cycle is about 203 days, which is 77 days longer than at Caterpillar. Liebherr holds its stock longer and pays its suppliers faster.

This is the biggest lever that management controls. Ten days less would release about 405 million euro. But this happens only once, so it cannot replace a better margin.



So what would have to change before an investment makes sense? We check three conditions on the 2026 statements.

• CLICK First, the margin should return to at least its four-year average of 3.9 per cent. In 2025 it was 2.8.

• CLICK Second, free cash flow should be positive and cover the distributions. In 2025 it was minus 176 million euro.

• CLICK Third, the investor needs information rights, for example on the 228 million euro of other costs that nobody can explain.

And 3.9 per cent is only a trigger, not a good return. The return on capital employed was 3.5 per cent before tax, against a benchmark of 7.7 per cent.

Also, a minority stake brings little influence, a low payout and shares that are hard to sell. So meeting the conditions starts a new analysis, not a purchase.

• CLICK – NEXT SLIDE



So here is our answer to the central question. Liebherr is financially strong, but operationally challenged.

We have three takeaways. First, the balance sheet is strong, and it survives our accounting tests. Second, returns and cash generation are too weak, and the gap to Caterpillar exists in every year. Third, the accounting analysis makes our verdict sharper.

If the margin stays near 2.8 per cent in 2026 and the net financial position turns negative, the investor should drop the case. In one sentence, see the turn in the figures before committing capital.

Thank you



Thank you.

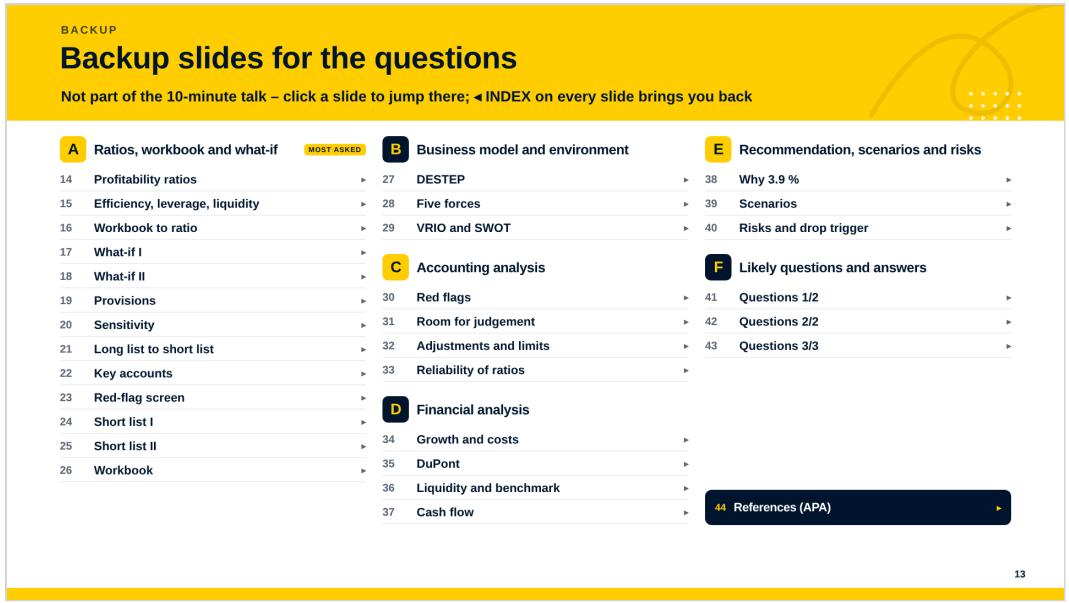
Simon Beermann · Adam Fliegel · Zofia Narkiewicz · Philipp Hoffmann · Zarah Gilmore Islam · HAT1 Financial Reporting · Fontys Venlo

Images: Liebherr-International AG

On behalf of the whole team, thank you very much for your attention. We are happy to answer your questions.

Q&A Press I for the backup index, every backup slide is one click away. Useful figures for questions. ROCE 2024 was 5.9 % before tax at a 4.6 % margin. The sustainable growth rate is 2.4 %. Personnel expenses were 28.6 % of revenue, against 26.4 % in 2023. Only €56 m of a €272 m profit was distributed in 2025. A deferred tax liability on Caterpillar's LIFO reserve at 21 % would lower its equity ratio only from 24.9 % to 24.2 %.

Backup index



Backup slides, not part of the 10-minute talk. Use them to answer questions: A covers ratios, formulas, what-if logic, provisions, sensitivities, the long and short list and the workbook – the most likely questions. B covers the business model, C the accounting analysis, D the financial analysis, E the recommendation and scenarios, F likely questions. Click a section to jump there; every backup slide links back to this index.

Backup A – Profitability ratios

BACKUP · A | RATIOS AND WORKBOOK

◂ INDEX

Profitability: returns are low because the margin is low

Margins and returns on capital – formula, Liebherr and Caterpillar, 2025

GROUP	RATIO	FORMULA	LIEBHERR	CATERPILLAR
PROFITABILITY margins	Operating margin	EBIT / revenue	2.8 %	17.2 %
	Pre-tax margin	Profit before tax / revenue	3.0 %	17.9 %
	Net profit margin	Profit after tax / revenue	1.8 %	13.8 %
	EBITDA margin	EBITDA / revenue	6.8 %	19.8 %
PROFITABILITY returns on capital	ROA	EBIT / average total assets	2.4 %	11.9 %
	ROCE	EBIT / (avg. equity + avg. non-current debt)	3.5 %	20.2 %
	ROE	Profit after tax / average equity	3.0 %	38.1 %*
	Pre-tax ROE	Profit before tax / average equity	4.8 %	49.4 %

* Not comparable – treasury shares reduce Caterpillar's equity. ROA is measured on EBIT, because Liebherr's interest expense is not disclosed.

DUPONT

ROE = net margin × asset turnover × leverage

leverage = average total assets / average equity

= **profitability** × **efficiency** × **leverage** – the ratio groups of slides 14 and 15

	MARGIN	TURNOVER	LEVERAGE	ROE
2025	1.84 %	× 0.85	× 1.89	3.0 %
pre-tax	2.96 %	× 0.85	× 1.89	4.8 %
2023 pre-tax	4.76 %	× 0.88	× 1.81	7.6 %

Leverage works on ROE, **not on the EBIT margin** – EBIT is measured before interest.

Source: Excel workbook, sheets D3 (blocks C and D) and D4; Report §5.3, Table J14. 2025; Caterpillar on the FIFO basis

14

We group the ratios into four families. This slide shows profitability, both margins and returns on capital. Return on equity is profit after tax divided by average equity, so 272 over 9,128, which is 3.0 per cent, or 4.8 per cent before tax. ROA, ROCE and ROE belong to profitability, because they measure profit against the capital used. DuPont links the families. ROE equals profitability times efficiency times leverage, and leverage is average total assets over average equity. Leverage works on ROE, not on the EBIT margin, because EBIT is measured before interest. Caterpillar's ROE is inflated by its treasury shares and is not comparable.

Backup A – Efficiency, leverage, liquidity

BACKUP · A | RATIOS AND WORKBOOK

Efficiency, leverage, liquidity: slow capital, solid finance

◂ INDEX

How capital is used, how it is financed and how liquid Liebherr is – 2025

GROUP	RATIO	FORMULA	LIEBHERR	CATERPILLAR
EFFICIENCY asset utilisation · working capital	Asset turnover	Revenue / average total assets	0.85	0.69
	Fixed asset turnover	Revenue / average PP&E	3.49	4.74
	Days inventory outstanding	Avg. inventories / revenue × 365	157.5	116.5
	Days sales outstanding	Avg. receivables / revenue × 365	73.8	54.5
	Days payables outstanding	Avg. payables / revenue × 365	28.6	44.9
	Cash conversion cycle	DIO + DSO − DPO	202.7	126.1
LEVERAGE solvency	Equity ratio	Equity / total assets	53.5 %	24.9 %
	Leverage (DuPont)	Avg. total assets / avg. equity	1.89	3.97
	Debt / equity	Debt incl. pensions / equity	0.40	1.84
	Debt / capital	Bank + lease liab. / (those + equity)	23.7 %	64.8 %
	Debt / EBITDA	Bank + lease liabilities / EBITDA	2.90	3.52
LIQUIDITY short term	Current ratio	Current assets / current liabilities	2.31	1.55
	Quick ratio	(Current assets − inventories) / current liab.	1.16	0.94
	Cash cover	Cash + current fin. assets / current debt	2.6×	–

READ WITH CARE

Days = efficiency, but they drive liquidity: 203 days of revenue must be financed.

Revenue basis: Liebherr reports no cost of sales; on its own cost basis DIO would be 299 days – not comparable.

Two debt measures: "debt" includes pensions (€3,714 m); bank and lease liabilities (€2,898 m) match the Damodaran benchmark.

Interest cover: not computable – no interest expense disclosed. Indicator: operating profit / interest paid 3.90.

PROFITABILITY
How much profit a company earns – on its revenue and on the capital it employs.

EFFICIENCY
How well a company uses its assets and working capital to generate revenue.

LEVERAGE
How strongly a company uses debt relative to its equity to finance its assets.

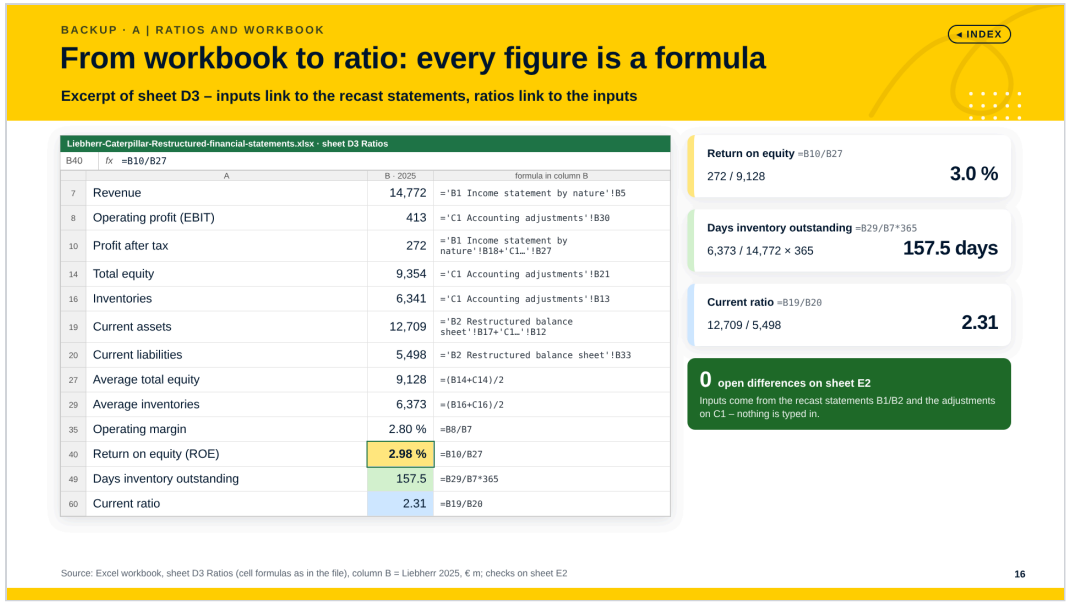
LIQUIDITY
Whether a company can pay its short-term obligations from its short-term assets.

Source: Excel workbook, sheets D3 (blocks D, E, G, H), D4 and D7; Report Tables J5 and J14. 2025; days on average balances; Caterpillar on the FIFO basis

15

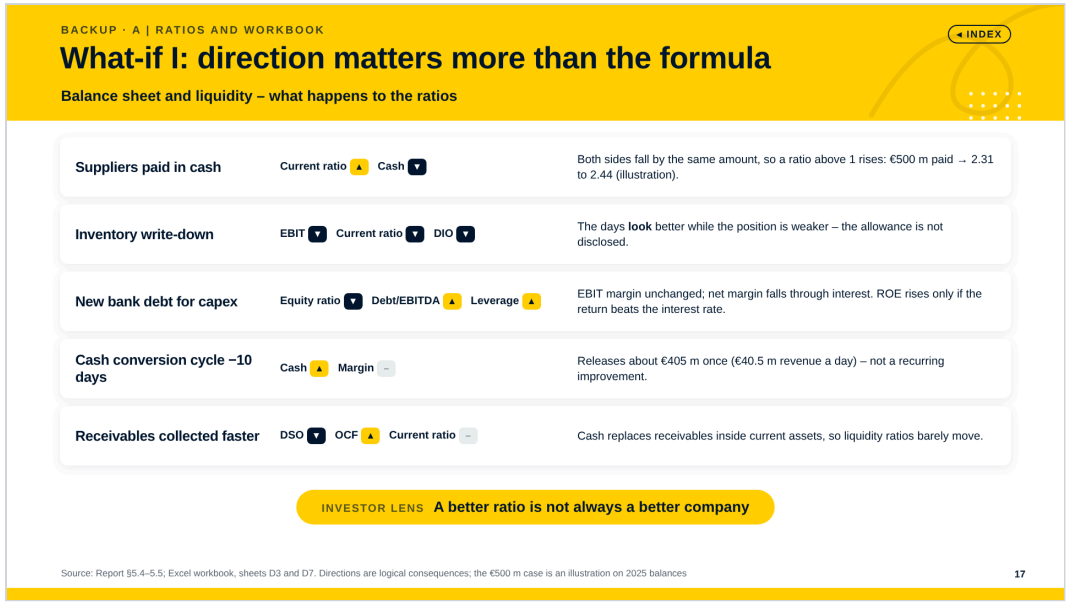
The other three families are efficiency, leverage and liquidity. Efficiency, also called asset utilisation, covers the turnover ratios and the working-capital days. The days are efficiency ratios, but they drive liquidity, because 203 days of revenue have to be financed. Leverage, or solvency, shows how Liebherr is financed in the long run, with an equity ratio of 53.5 per cent and a DuPont leverage of 1.89 against 3.97 at Caterpillar. Liquidity is short term. The current ratio of 2.31 and the quick ratio of 1.16 are liquidity ratios, not leverage ratios. Days are on a revenue basis because Liebherr reports no cost of sales, and there are two debt measures, one including pensions of 3,714 million euro and the bank and lease liabilities of 2,898 million euro used for the benchmark.

Backup A – Workbook to ratio



This is an excerpt of sheet D3 for Liebherr in 2025. Every input is a formula that links to the recast statements or to the adjustments sheet, so nothing is typed in. Return on equity in cell B40 is B10 divided by B27, so 272 over the average equity of 9,128, which gives 3.0 per cent. Inventory days are average inventories divided by revenue times 365, which gives 157.5 days. The current ratio is current assets of 12,709 divided by current liabilities of 5,498, which gives 2.31. Sheet E2 reconciles every subtotal, with zero open differences.

Backup A – What-if I



If suppliers are paid in cash, current assets and current liabilities fall by the same amount, so a current ratio above one rises – with 500 million euro it would go from 2.31 to 2.44. An inventory write-down lowers profit and the current ratio, but also the inventory days, so the ratio looks better while the position is weaker. New debt lowers the equity ratio and raises leverage, but leaves the EBIT margin unchanged. A shorter cash conversion cycle releases cash once, it does not improve the margin.

Backup A – What-if II

BACKUP · A | RATIOS AND WORKBOOK

What-if II: accounting choices move profit, not cash

◂ INDEX

Effect of an accounting choice on EBIT, equity and cash

CHANGE	EBIT	EQUITY	CASH / OCF	LIEBHERR EXAMPLE
Provision addition	▼	▼	–	Expense now – the obligation is estimated
Provision usage (utilisation)	–	–	▼	Obligation settled – no profit effect
Provision reversal (release)	▲	▲	–	Estimate no longer needed – profit without cash
Slower depreciation (longer life)	▲	▲	–	Liebherr 5.5 % vs. Caterpillar 6.6 % of cost
More own work capitalised	▲ now	▲	–	€317 m = 77 % of 2025 EBIT; returns as depreciation
LIFO → FIFO, rising prices (CAT)	▲	▲	–	+$441 m EBIT, +$4,305 m equity before tax

INVESTOR LENS **Operating cash flow is the check on the accruals – only 28.4 % of EBITDA in 2022–2025**

Source: Report §4.2–4.4 and §6.2; Excel workbook, sheets A2 (note 10), C1, C2 and D9

18

Accounting choices move profit, not cash. A provision addition is an expense, usage settles the obligation in cash without a profit effect, and a reversal increases profit without cash. Slower depreciation and more own work capitalised raise EBIT today. Caterpillar's switch from LIFO to FIFO raises profit and equity. That is why operating cash flow, only 28.4 per cent of EBITDA over four years, is our check on the accruals.

Backup A – Provisions

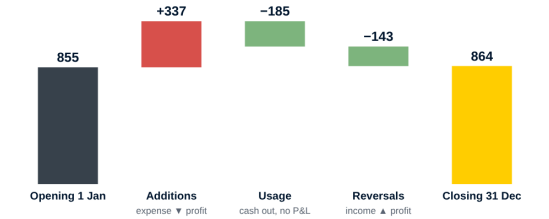
BACKUP · A | RATIOS AND WORKBOOK

Provisions: reversals rose – but so did additions

◂ INDEX

Movement of provisions 2025 (IAS 37), € m – of which warranty 530

Roll-forward 2025, all provisions



Net charge = additions − reversals	2022	2023	2024	2025
	193	338	195	194

How to read it

- **IAS 37:** present obligation from a past event, probable outflow, reliable estimate
- **Usage ≠ reversal:** only additions and reversals touch profit
- **Hidden reserve test:** a release to support profit would show as a *falling* net charge – 195 → 194, so not visible
- **Flag kept:** other provisions built up by €146 m in 2023 are being released (€63 m in 2025); the notes cannot show why

Source: Excel workbook, sheets A2 (note 10, rows 68–80), C2 and D11 (block B); Report §4.2 and §4.4, Table H1 test 1. € m

19

Provisions under IAS 37 need a present obligation from a past event, a probable outflow and a reliable estimate. In 2025 the opening balance of 855 million euro rose by 337 of additions, fell by 185 of usage and by 143 of reversals, to 864. Usage is the real outflow and has no profit effect; only additions and reversals touch profit. The net charge – additions less reversals – stayed at 194 against 195, so a hidden reserve being released to support profit is not visible. We keep the flag because other provisions built up in 2023 are now being released and the notes cannot explain why.

Backup A – Sensitivity

BACKUP · A | RATIOS AND WORKBOOK

Sensitivity: every measured effect lowers the margin

◂ INDEX

Liebherr operating margin 2025 – illustrations on published figures, not booked adjustments

STEP	CALCULATION, € M	EBIT	MARGIN
As reported	–	413	2.80 %
Net provision charge at the 2022–2024 average	413 - (1.76 % - 1.31 %) × 14,772	347	2.35 %
… without the high charge of 2023	413 - (1.43 % - 1.31 %) × 14,772	395	2.68 %
Caterpillar's depreciation rate (6.6 % instead of 5.5 % of cost)	413 - 105	308	2.08 %
Both effects together	413 - 66 - 105	242	1.64 %

2.80 − 0.44 − 0.71 = 1.64 %

Sheet D9 also shows a **gross-reversal** view (2.5 % at average reversals, 1.8 % without any reversal). The report uses the **net charge** (D11), because additions rose together with the reversals. Own work capitalised (€317 m) cannot be measured.

Source: Excel workbook, sheets D9 (blocks A and B) and D11 (block B); Report §6.1, Tables H3 and J13

20

We measure two effects without booking them. If the net provision charge had been at its 2022 to 2024 average of 1.76 per cent of revenue instead of 1.31, EBIT would be 347 million euro and the margin 2.35 per cent; without the high charge of 2023 it would be 2.68. At Caterpillar's depreciation rate depreciation would be 105 million euro higher and the margin 2.08 per cent. Both together give 242 million euro, or 1.64 per cent. Sheet D9 also shows a gross-reversal view, but the report uses the net charge because additions rose with the reversals.

Backup A – Long list to short list



Step one is the long list of 30 accounting treatments with room for choice. Step two keeps the nine that combine a key account with high or medium-to-high flexibility. Step three is the red-flag screen: eight tests plus one consistency check, with four flags, three clean and two not testable. Step four is the short list of 22 findings: seven that no adjustment can remove, five reversed in the workbook and ten the disclosure does not allow us to reverse. The two counts of nine are a coincidence.

Backup A – Key accounts

BACKUP · A | RATIOS AND WORKBOOK

Seven key accounts carry the judgement

Key accounts × flexibility = the nine key treatments of the long list

◂ INDEX

KEY ACCOUNT	2025, € M	STANDARD AND JUDGEMENT	FLEXIBILITY	WHY IT IS ON THE SHORT LIST
Inventories long list B1, B2	6,341 42.9 % of revenue	**IAS 2** – Fixed overheads absorbed at an undefined normal capacity; allowance not disclosed	High	Largest account – 36.3 % of total assets; quality cannot be verified
Provisions long list C1–C3	864 of which warranty 530	**IAS 37** – Recognition, measurement and reversal of the estimate	High	Reversals €143 m = 34.6 % of the operating result if credited within it
PP&E and rental fleet long list B5–B7	4,304 24.6 % of total assets	**IAS 16 / IAS 36** – Useful lives, residual values, indicator-based impairment	High (impairment), medium (lives)	Depreciation 5.5 % of cost vs. 6.6 % at Caterpillar
Own work capitalised long list B3	317 77 % of the operating result	**IAS 16 / IAS 38** – Which production cost is capitalised rather than expensed	Medium to high	Moves cost out of the year – returns later as depreciation
Trade receivables and contract assets long list B11	3,116 17.8 % of total assets	**IFRS 9** – Simplified lifetime expected credit loss through a provision matrix	Medium	Grew 9.1 % against revenue +1.0 % in 2025; allowance not disclosed
Revenue long list A1, A2	14,772	**IFRS 15** – Point in time against over time; cost-to-cost estimate	Medium	Share recognised over time not disclosed – affects every ratio
Deferred tax assets and tax charge long list D1	285 tax rate 37.8–99.2 %	**IAS 12** – Probability of future taxable profit; announced instead of enacted rates	High	No tax reconciliation – after-tax ratios not usable

Source: Report §4.2–4.3, Tables G1 and G2; Liebherr-International AG (2026); IFRS Foundation (n.d.)

22

Seven key accounts carry the judgement. Inventories under IAS 2, provisions under IAS 37, property, plant and equipment under IAS 16 and 36, own work capitalised, receivables under IFRS 9, revenue under IFRS 15 and deferred taxes under IAS 12. They are on the short list because they combine a material amount with high or medium flexibility: inventories are 36.3 per cent of total assets, provision reversals equal 34.6 per cent of the operating result, own work capitalised 77 per cent, and the tax rate swings between 37.8 and 99.2 per cent without a reconciliation.

Backup A – Red-flag screen

BACKUP · A | RATIOS AND WORKBOOK

Red-flag screen: nine tests, four flags

4 raised · 3 clean · 2 not testable – reasons to ask, not proof of manipulation

◂ INDEX

#	TEST	EVIDENCE 2022–2025	VERDICT
1	**Unexplained transactions that boost profits**	Provision reversals 90 → 143 (2022–2025) = 34.6 % of 2025 EBIT; net charge flat (195 → 194); credited line not disclosed	Raised
2	**Inventories or receivables rise faster than sales**	Inventories +27.3 % vs. revenue +8.2 % (2022); receivables +9.1 % vs. revenue +1.0 % (2025)	Raised twice
3	**Gap between profit and cash flow or taxable profit**	Operating cash flow −415 and −10 (2022–23) against profit; tax rate 99.2 % → 37.8 %, no reconciliation	Raised
4	**R&D partnerships, special purpose entities, sold receivables**	None disclosed in four years; R&D fully expensed	No flag
5	**Unexpected large asset write-offs**	PP&E impairment 8, 55, 9, 3 – 2023 six times the other years, same one-sentence explanation	Raised for 2023
6	**Large year-end adjustments**	No interim figures published, so no within-year comparison	Not testable
7	**Qualified opinions or auditor changes**	Ernst & Young all four years, no qualification, no restatement (ISA 810 opinion)	No flag
8	**Related party transactions**	No note in four years – family-owned group that also runs six hotels	Not testable
9	**Changes in accounting policies or estimates**	Policies worded identically; new standards without material effect	No flag

Source: Report §4.4, Table H1 (8 tests of step 5 + consistency check of step 3); Liebherr-International AG (2023–2026)

We ran the eight red-flag tests of step five over four years, plus one consistency check of step three. Four raised a flag. Provision reversals rose while the result fell, inventories and receivables grew faster than sales, the tax rate swung without a reconciliation, and the impairment of 2023 was six times the normal level. Three tests are clean, namely off-balance-sheet structures, the audit opinion and policy changes. Two cannot be run, because there are no interim figures and no related party note. A red flag is a reason to ask, not proof of manipulation.

Backup A – Short list I

BACKUP · A | RATIOS AND WORKBOOK

Short list I: what cannot be fixed – and what we fixed

◂ INDEX

Findings 1–12 of 22 (Table H2)

#	FINDING	ACCOUNT AFFECTED (€ M / $ M)	TREATMENT
NOT ADJUSTABLE – RED FLAGS AND DISCLOSURE GAPS · 7			
1	**Provision reversals rise as the result falls (test 1)**	Provisions, 864	Not adjustable: the income line carrying the reversal is not disclosed
2	**Inventories and receivables grow faster than sales (test 2)**	Inventories 6,341, receivables 3,116	Not adjustable: neither allowance is disclosed
3	**Volatile and unreconciled tax charge (test 3)**	Deferred tax assets, 285 (IAS 12)	Not adjustable: no tax reconciliation is published
4	**Impairment spike in 2023 (test 5)**	PP&E, impairment 55	Not adjustable: no test assumptions are disclosed
5	**Composition of other operating expenses not disclosed**	Other operating expenses, 2,170	Not adjustable: the line has no note
6	**No related party disclosure in four years (test 8)**	Whole group	The information is not in the published statements
7	**Customer credit risk carried off balance sheet**	Financial guarantees, not quantified	Not adjustable: no amount is disclosed
REVERSED IN THE WORKBOOK · 5			
8	**Caterpillar inventories on LIFO**	Inventories, equity, operating profit	FIFO excess disclosed – exact correction: $4,305 m
9	**Caterpillar goodwill impairment 2022 inside operating costs**	Operating profit 2022	Moved to other income as a non-recurring item
10	**Liebherr change in inventories and own work shown as income**	Revenue, cost of materials	Both mapped into the cost line (standardised format)
11	**Caterpillar finance receivables inside trade receivables**	Trade receivables	Mapped to other current assets and non-operating investments
12	**Liebherr trade payables not shown on the balance sheet**	Trade payables	Taken from note 7, published subtotal preserved

Source: Report §4.4, Table H2 (short list, 22 findings); Excel workbook, sheets B3, C1, C2, E1 and E2

24

The short list has 22 findings. Seven are red flags or disclosure gaps that no adjustment can remove – for example the provision reversals, the unexplained tax charge and the other operating expenses without a note. Five distortions we reversed in the workbook – the most important is Caterpillar's LIFO inventory, restated to FIFO with the disclosed excess of 4,305 million dollars.

Backup A – Short list II

BACKUP · A | RATIOS AND WORKBOOK

Short list II: what the disclosure does not let us reverse

◂ INDEX

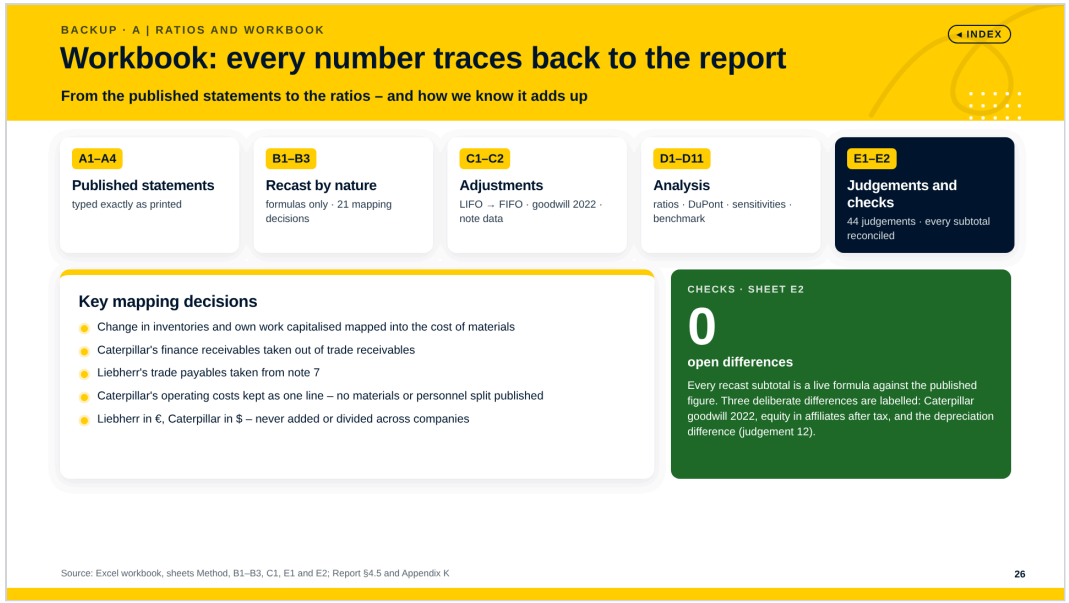
Findings 13–22 of 22 (Table H2)

#	FINDING	ACCOUNT AFFECTED (€ M / $ M)	TREATMENT
NOT REVERSIBLE – DISCLOSURE TOO THIN · 10			
13	**Gross presentation of rental machine disposals**	Revenue, cost of materials	The amount is not disclosed
14	**The interest element of the finance result**	Finance income 1,043, expenses 1,015	No split is published
15	**No segment note under IFRS 8**	Whole income statement	Segment data only in the management report
16	**The effect of hyperinflation accounting**	Turkish and Ghanaian entities	Not quantified in any year
17	**Actuarial assumptions behind the pension obligation**	Post-employment benefits, 791	No discount rate and no sensitivity disclosed
18	**Inventory and receivable allowances**	Inventories 6,341, receivables 3,116	Neither balance nor charge is disclosed
19	**The two groups do not depreciate at the same rate**	PP&E, 4,304	Measured (Table H3), not carried into the recast – asset mixes differ
20	**Cat Financial inside every Caterpillar figure**	Caterpillar balance sheet	The consolidated basis was set by the assignment
21	**Caterpillar discloses no cost of materials or personnel expense**	Cost structure comparison	Not estimated – operating costs shown as one line
22	**Potential refund of US tariffs after the balance sheet date**	Not recognised; note 12	Non-adjusting event (IAS 10), no reliable estimate

Source: Report §4.4, Table H2 (short list, 22 findings); Excel workbook, sheets B3, C1, C2, E1 and E2

Ten distortions cannot be reversed because the disclosure is too thin: for example the gross presentation of rental machine disposals, the missing interest split, the missing segment note, the pension assumptions and the undisclosed allowances. The different depreciation rates we measured in a sensitivity, but did not book, because the asset mixes differ.

Backup A – Workbook



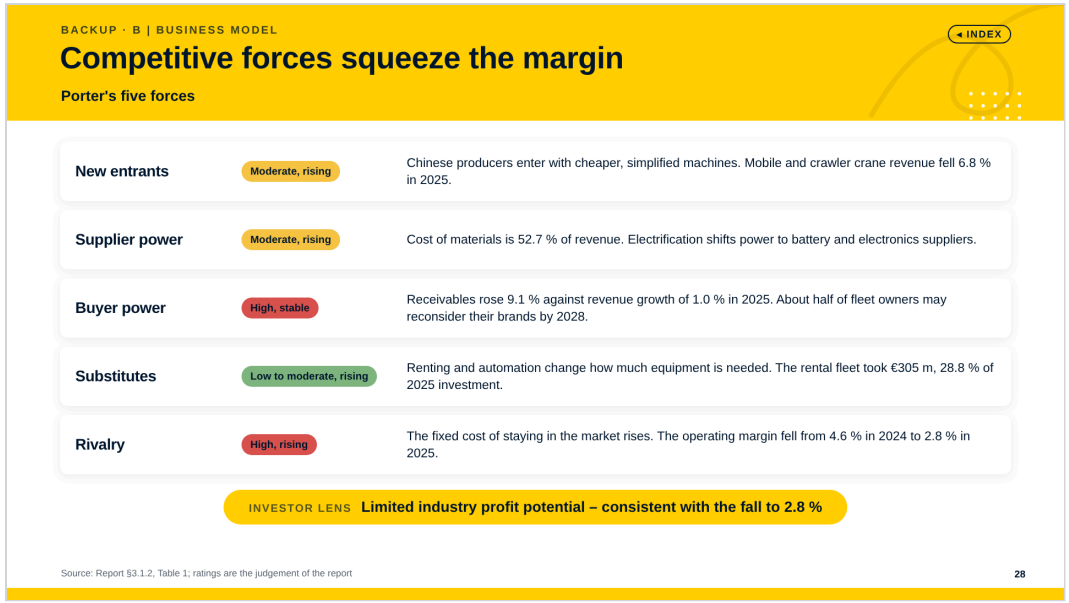
The workbook starts with the published statements typed exactly as printed. The B sheets recast both companies into one format by nature using formulas only, with 21 documented mapping decisions. The C sheets hold the LIFO to FIFO restatement and the note data, the D sheets the analysis. Sheet E1 documents 44 judgements and sheet E2 reconciles every recast subtotal with the published figure: zero open differences, apart from three labelled ones.

Backup B – DESTEP



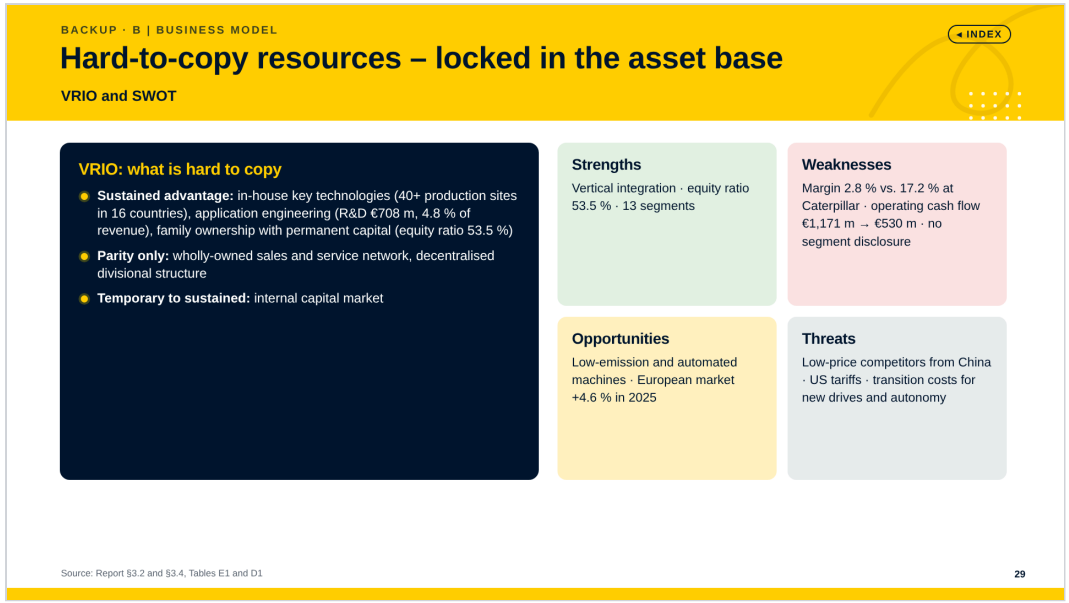
The DESTEP scan shows a modest European recovery from 2026, set against higher financing costs, tariff uncertainty and a technology and labour transition that raises costs before it raises prices. Economic and political factors carry the largest near-term risk, technological and ecological factors the largest long-term opportunity. A market recovery alone does not show that the margin recovers.

Backup B – Five forces



Buyer power and rivalry are high, the threat of new entrants and supplier power are moderate and rising, and substitutes are a low to moderate threat. The industry profit potential is therefore limited and squeezed from the customer side and by low-cost entrants. This is consistent with the fall of the operating margin to 2.8 per cent in 2025 against 17.2 per cent at Caterpillar.

Backup B – VRIO and SWOT



The VRIO test shows that in-house technology, application engineering and family capital are sustained advantages, while the sales and service network is only at parity, because Caterpillar's independent dealers provide comparable access. The SWOT sums it up: vertical integration and the equity ratio are the strengths that matter, the weakness is the margin and the weakening internal funding, the opportunity is demand for low-emission and automated machines, and the threats are low-price competitors and tariffs.

Backup C – Red flags

BACKUP · C | ACCOUNTING ANALYSIS

Red-flag screen: reasons to ask, not proof

◂ INDEX

Nine tests over four years

Four red flags

- **Provision reversals:** €143 m in 2025, 34.6 % of the operating result if credited within it
- **Receivables vs. revenue:** receivables +9.1 % against revenue +1.0 % in 2025
- **Profit vs. tax:** effective tax rate 99.2 %, 45.1 %, 58.2 % and 37.8 % (2022–2025), no reconciliation
- **Impairment:** €55 m in 2023, six times the level of the other years

Clean or not testable

- **Clean:** no off-balance-sheet arrangements disclosed · same auditor (Ernst & Young), no qualified opinion · no change of policy or estimate
- **Not testable:** related party transactions (no note in four years) · large year-end adjustments (no interim figures)

4
flags raised

3
clean

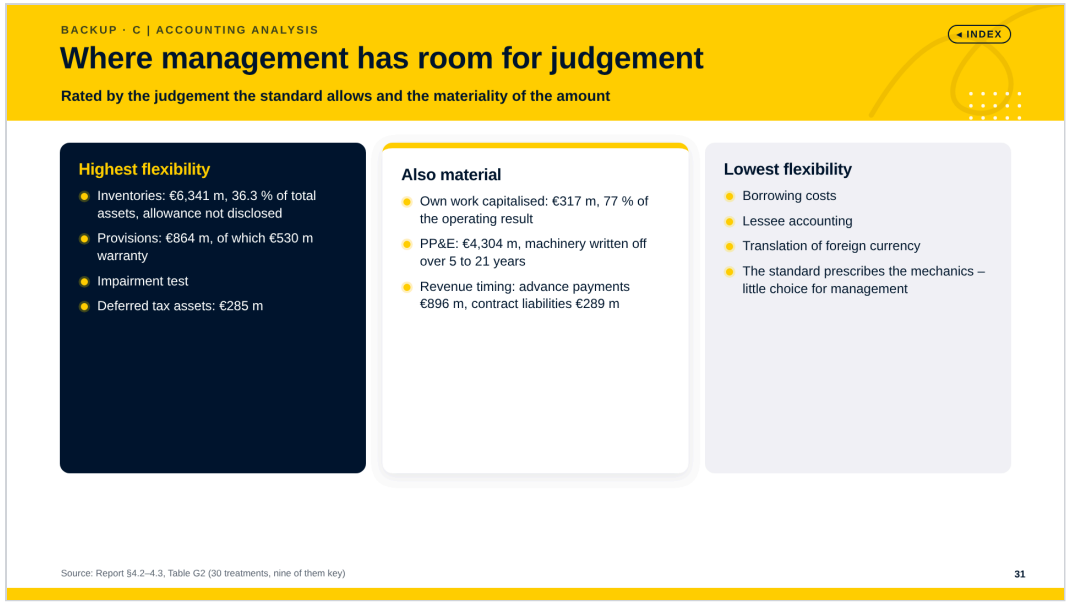
2
not testable

Source: Report §4.4, Table H1; financial statements 2022–2025

30

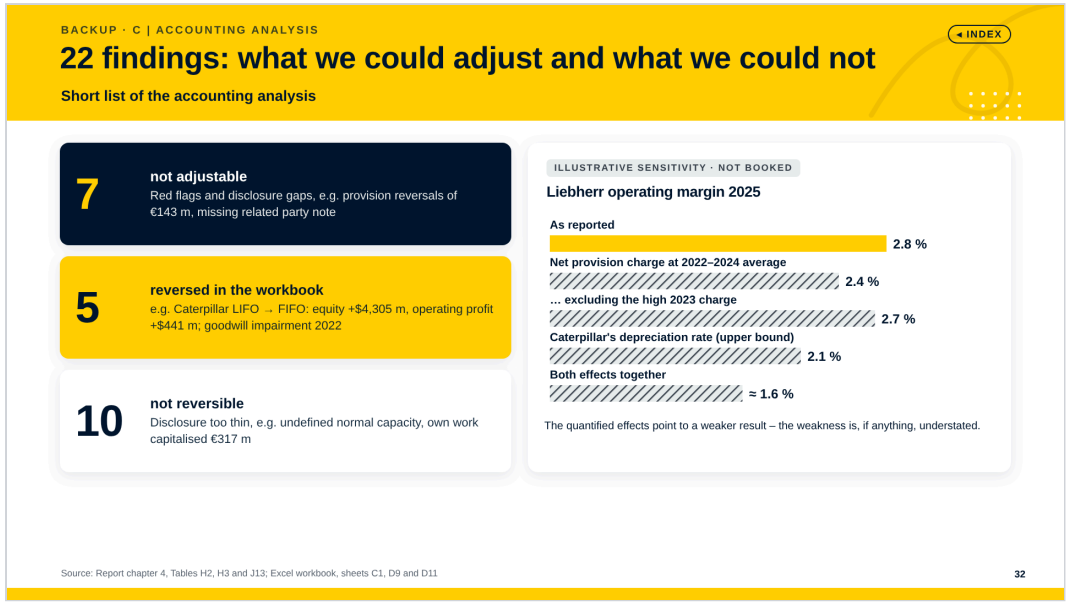
We ran nine red-flag tests over four years. Four raised a flag, three came back clean and two cannot be run. The flags are the provision reversals of 143 million euro, receivables growing faster than revenue, an effective tax rate between 37.8 and 99.2 per cent without a reconciliation, and the impairment of 55 million euro in 2023. The tests that cannot be run are related party transactions and year-end adjustments. Red flags are reasons to ask, not proof of manipulation.

Backup C – Room for judgement



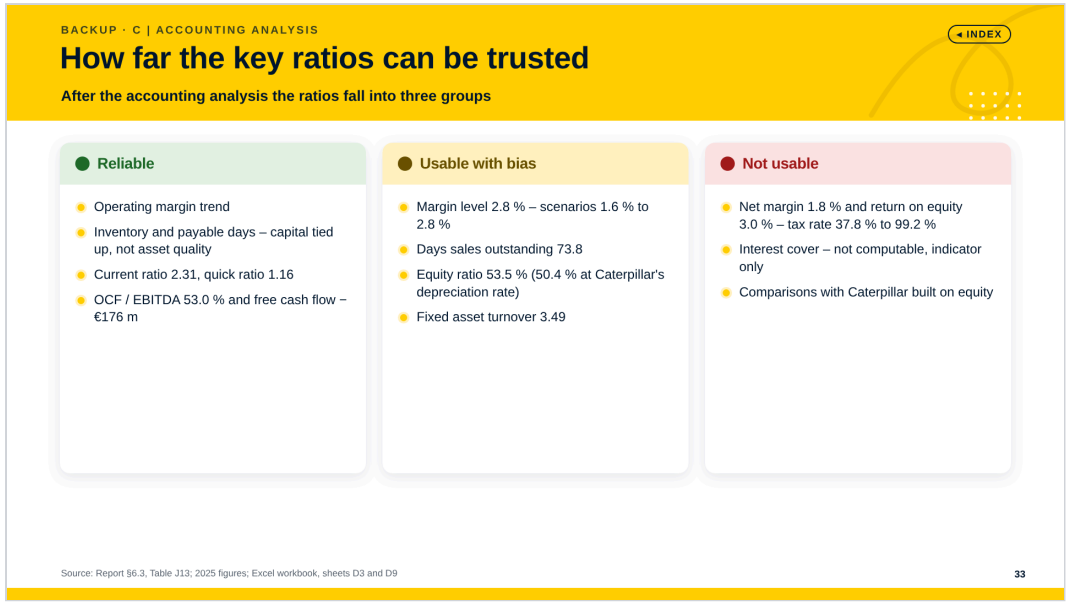
Flexibility is rated on two criteria: the degree of judgement the standard permits and the materiality of the amount. It is highest for inventories and their allowance, the impairment test, the provisions and deferred tax assets. Own work capitalised, property, plant and equipment and revenue timing follow. It is lowest where the standard prescribes the mechanics, as with borrowing costs, lessee accounting and currency translation.

Backup C – Adjustments and limits



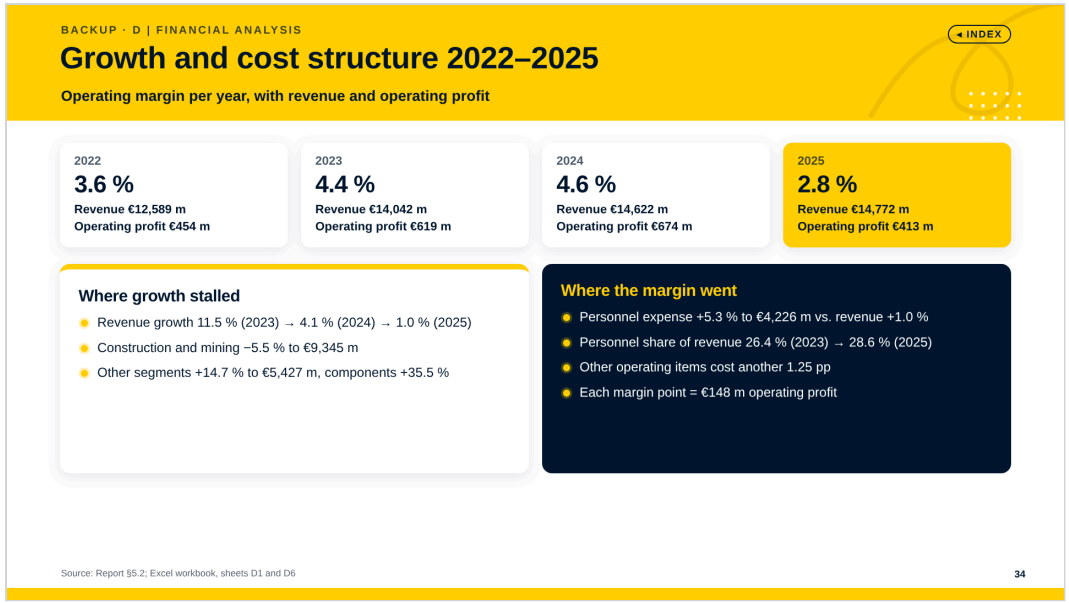
The short list holds 22 accounting findings. Seven are red flags or disclosure gaps that no adjustment can remove, five were reversed in the workbook, and ten cannot be reversed because the disclosure is too thin. The quantified effects make the 2025 result look better, not worse: the operating margin of 2.8 per cent falls to between about 1.6 and 2.7 per cent in the measured scenarios, so the weakness is, if anything, understated.

Backup C – Reliability of ratios



After the accounting analysis the ratios fall into three groups. Reliable are the trend of the operating margin, inventory and payable days, liquidity and cash conversion. Usable with a stated bias are the level of the margin, days sales outstanding, fixed asset turnover and the equity ratio. Not usable for a conclusion are the net margin and the return on equity as a trend, interest cover and every comparison with Caterpillar built on equity.

Backup D – Growth and costs



Revenue growth slowed from 11.5 per cent in 2023 to 1.0 per cent in 2025. The slowdown came from construction and mining, which fell 5.5 per cent, while the other segments grew 14.7 per cent. The margin fell from 4.6 to 2.8 per cent because personnel expense grew 5.3 per cent against revenue growth of 1.0 per cent, and other operating items added to the burden. Each margin point is worth 148 million euro.

Backup D – DuPont

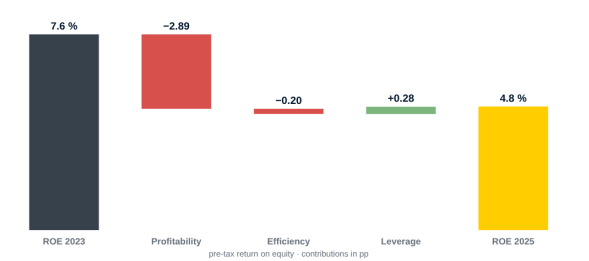
BACKUP · D | FINANCIAL ANALYSIS

DuPont: the margin, not leverage, drives returns down

◂ INDEX

Pre-tax return on equity 2023 → 2025

Contributions to the change in pre-tax ROE, percentage points



What it means

- **Profitability −2.89 pp:** lower pre-tax margin
- **Efficiency −0.20 pp:** slightly lower asset turnover
- **Leverage +0.28 pp:** more debt even helped a little
- After tax, ROE was 3.0 % vs. 16.4 % in the US industry – but the tax charge is volatile, so the trend is read before tax

Source: Report §5.3, Figure 2a, Table J4; Excel workbook, sheets D4 and D6 (logarithmic split)

35

We read the return on equity before tax because the tax charge is volatile. The pre-tax return fell from 7.6 per cent in 2023 to 4.8 per cent in 2025. The DuPont decomposition assigns 2.89 percentage points of the fall to the lower pre-tax margin and 0.20 points to lower asset turnover, while higher leverage added 0.28 points. Leverage is therefore not the cause – the margin is.

Backup D – Liquidity and benchmark

BACKUP · D | FINANCIAL ANALYSIS

Strong solvency, far weaker returns

Liquidity, solvency and benchmark 2025

◂ INDEX

Balance sheet strength

- Equity ratio 53.5 %; bank and lease liabilities 23.7 % of capital (industry 39.3 %)
- Cash and current financial assets €3,147 m cover current debt of €1,215 m 2.6×
- Current ratio 2.31 vs. 1.55 at Caterpillar
- Net financial position €523 m (2022) → net debt €85 m (2023) → €249 m (2025)
- Debt / EBITDA 2.14 → 2.90; operating profit / interest paid (indicator) 11.07 (2022) → 3.90 (2025)

2025	LIEBHERR	US INDUSTRY	CATERPILLAR
Operating margin	**2.8 %**	15.9 %	17.2 %
EBITDA margin	**6.8 %**	19.6 %	19.8 %
Return on equity	**3.0 %**	16.4 %	38.1 %*
Inventories, % of revenue	**42.9 %**	16.6 %	33.2 %
Debt / EBITDA	**2.90**	2.30	3.52

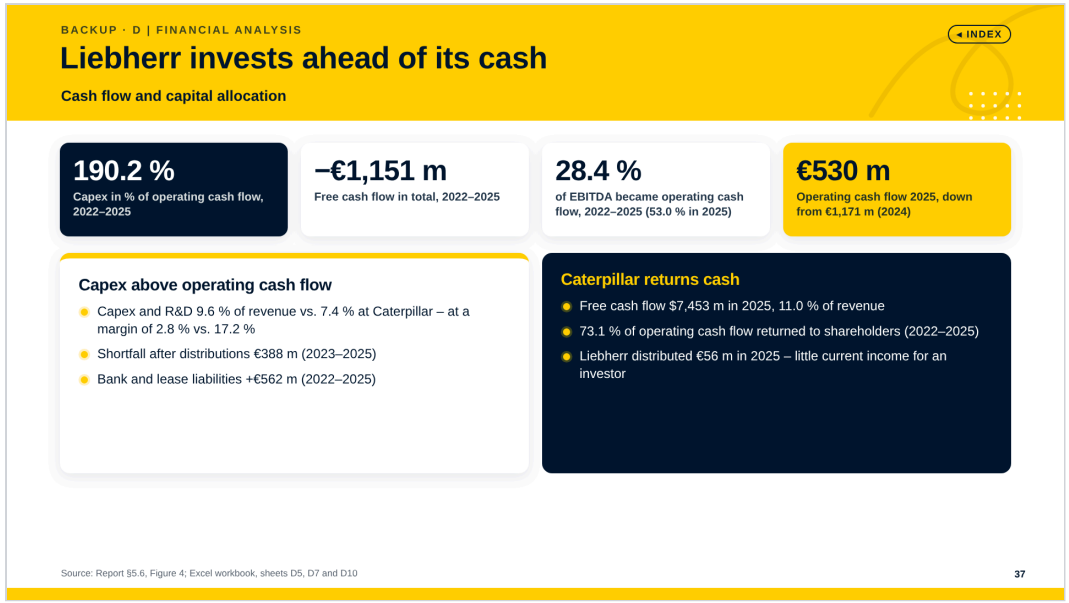
* Not comparable – Caterpillar's equity is reduced by treasury shares. Caterpillar on the FIFO basis.

Source: Report §5.1 and §5.5, Table J14; Damodaran (2026b, 2026c, 2026d, 2026e), US machinery industry, January 2026; Excel workbook, sheets D3, D7 and D10

36

Solvency is the clearest strength, with an equity ratio of 53.5 per cent and a current ratio of 2.31. The cushion is shrinking, though: the net financial position fell from 523 million euro in 2022 to a net debt of 85 million in 2023 and stood at 249 million in 2025, and debt to EBITDA rose to 2.90, above the industry's 2.30. Against the industry and Caterpillar, margins and returns are far lower and inventories tie up much more capital. Caterpillar's return on equity is not comparable because of its treasury shares.

Backup D – Cash flow



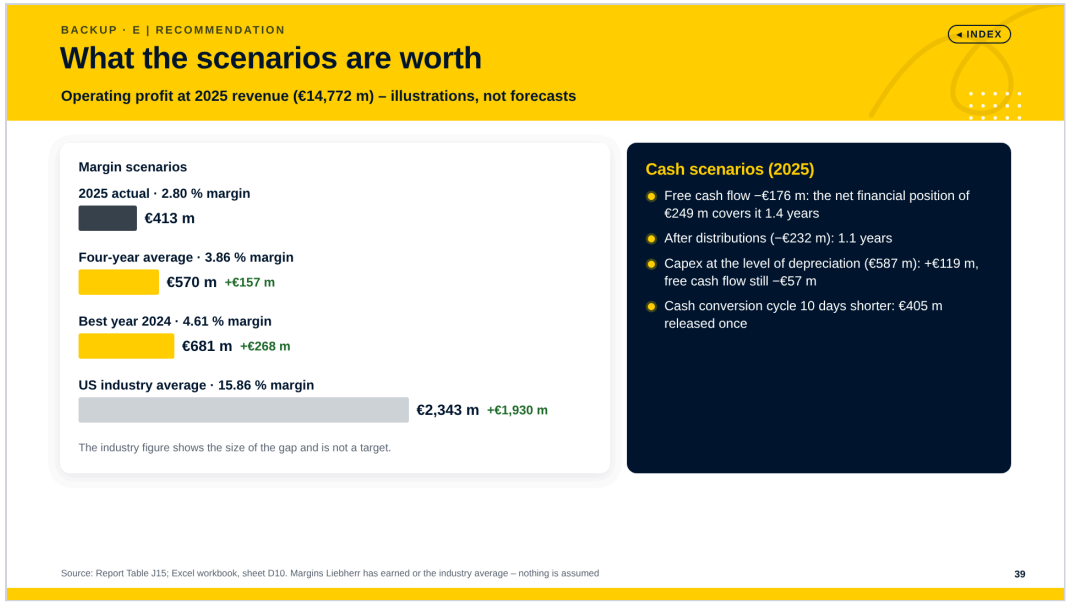
Cash flow is the weakest part of the figures. From 2022 to 2025 capital expenditure equalled 190.2 per cent of operating cash flow and free cash flow totalled minus 1,151 million euro. Operating cash flow fell from 1,171 million euro in 2024 to 530 million in 2025. Liebherr spends more than Caterpillar relative to revenue but earns a far lower margin, and borrowing helped fund investment and distributions. Caterpillar generated free cash flow of 7,453 million dollars and returned 73.1 per cent of its operating cash flow to shareholders.

Backup E – Why 3.9 %



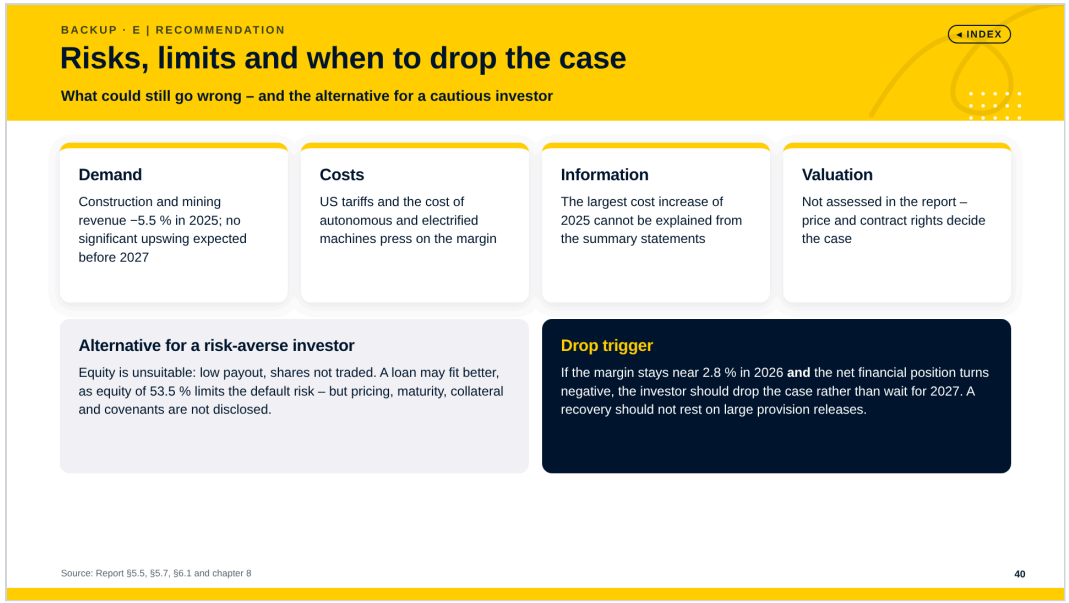
The margin of 3.9 per cent is the four-year average and therefore a realistic first test of recovery. At 2025 revenue it would add 157 million euro of operating profit. It is a trigger, not a target return: even in 2024, at a margin of 4.6 per cent, the return on capital employed was 5.9 per cent before tax, below the 7.70 per cent after-tax benchmark. Earning the benchmark would need a margin of about 6 to 8 per cent. The benchmark is that of listed US machinery companies, not Liebherr's own cost of capital.

Backup E – Scenarios



The margin scenarios apply margins Liebherr has already earned to the revenue of 2025 – illustrations on published figures, not forecasts. The industry margin shows the size of the gap and is not a target. On the cash side the net financial position of 249 million euro covers the free cash flow shortfall of 176 million euro for 1.4 years and the shortfall after distributions for 1.1 years. Capital expenditure at the level of depreciation would improve free cash flow by 119 million euro, but it would still be negative at minus 57 million. Ten fewer days in the cash conversion cycle would release about 405 million euro, once.

Backup E – Risks and drop trigger



Demand, tariffs and the cost of new technology are the main external risks. Financing is not the immediate constraint, but prolonged underperformance would make it one. The investor needs information rights because the largest cost increase of 2025 cannot be explained. The report does not assess a valuation. For a risk-averse investor an equity stake is unsuitable and a loan might fit better in principle, but its terms are not disclosed. The drop trigger is a margin near 2.8 per cent in 2026 together with a negative net financial position.

Backup F – Questions 1/2

BACKUP · F | LIKELY QUESTIONS

Why not invest now?

Likely questions (1/2)

◂ INDEX

Q Why not invest now?	Neither financial condition was met in 2025: the margin of 2.8 % is below its four-year average of 3.9 %, and free cash flow was −€176 m. The answer is **not yet**, rather than no.
Q Why is 3.9 % the first test?	It is the average Liebherr earned from 2022 to 2025 – a realistic first sign of recovery. At 2025 revenue it adds €157 m of operating profit. A trigger, not a target return.
Q Is the LIFO to FIFO adjustment fair?	IAS 2 does not permit LIFO, so FIFO puts both groups on one basis. Caterpillar's LIFO reserve of $4,305 m lifts its margin only from 16.50 % to 17.15 % – the gap stays large.

Source: Report chapters 4 to 8; Excel workbook, sheets C1, D3 and D10

41

Answer 1 is the recommendation in short. Answer 2 explains why the four-year average is only a first test. Answer 3 shows that the adjustment of Caterpillar to FIFO is small compared with the gap in margin.

Backup F – Questions 2/2

BACKUP · F | LIKELY QUESTIONS

What the balance sheet cannot fix

Likely questions (2/2)

◂ INDEX

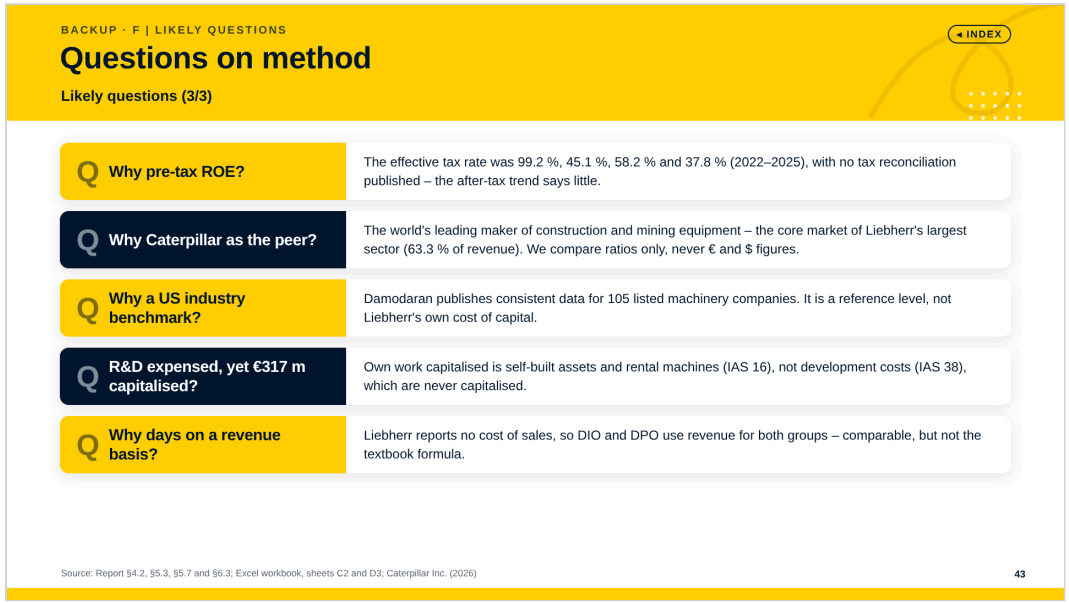
Q What do the red flags mean?	They are reasons to ask, not proof of manipulation. Four flags are raised, three tests are clean and two cannot be run on the published statements – that is why information rights come first.
Q Why is ROCE so low?	ROCE is 3.5 % before tax against a 7.70 % after-tax benchmark: the margin is only 2.8 %, and inventories of 42.9 % of revenue and a cash cycle 77 days longer than Caterpillar's tie up capital.
Q Why is a strong balance sheet not a return?	An equity ratio of 53.5 % and a net financial position of €249 m limit the risk, but do not earn the cost of capital. They would cover a shortfall like 2025 (−€232 m) for about 1.1 years.

Source: Report chapters 4 to 8; Excel workbook, sheets C1, D3 and D10

42

The red flags are reasons to ask. ROCE is low because of the low margin and the capital tied up in inventories. The balance sheet protects the investor, but it cannot replace better returns and cash conversion.

Backup F – Questions 3/3



We read returns before tax because the effective tax rate swung between 37.8 and 99.2 per cent without a reconciliation. Caterpillar is the world's leading maker of construction and mining equipment, the market of Liebherr's largest sector, and we only compare ratios. The Damodaran industry is a reference level, not Liebherr's cost of capital. Own work capitalised is self-built equipment, not development costs. Working-capital days use revenue because Liebherr reports no cost of sales.

BACKUP · REFERENCES

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◂ INDEX

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Own work: Beermann, S., Fliegel, A., Narkiewicz, Z., Hoffmann, P., & Gilmore Islam, Z. (2026). *Liebherr Group financial report: Investment analysis 2022–2025* and the Excel workbook *Liebherr–Caterpillar restructured financial statements*. Fontys University of Applied Sciences, Venlo. · Images: Liebherr-International AG. · Full entries with URLs: report, References.

44

Reference list for every source cited on the slides, in APA 7th edition. The full entries with URLs are in the report. Use this slide if someone asks where an external figure comes from.